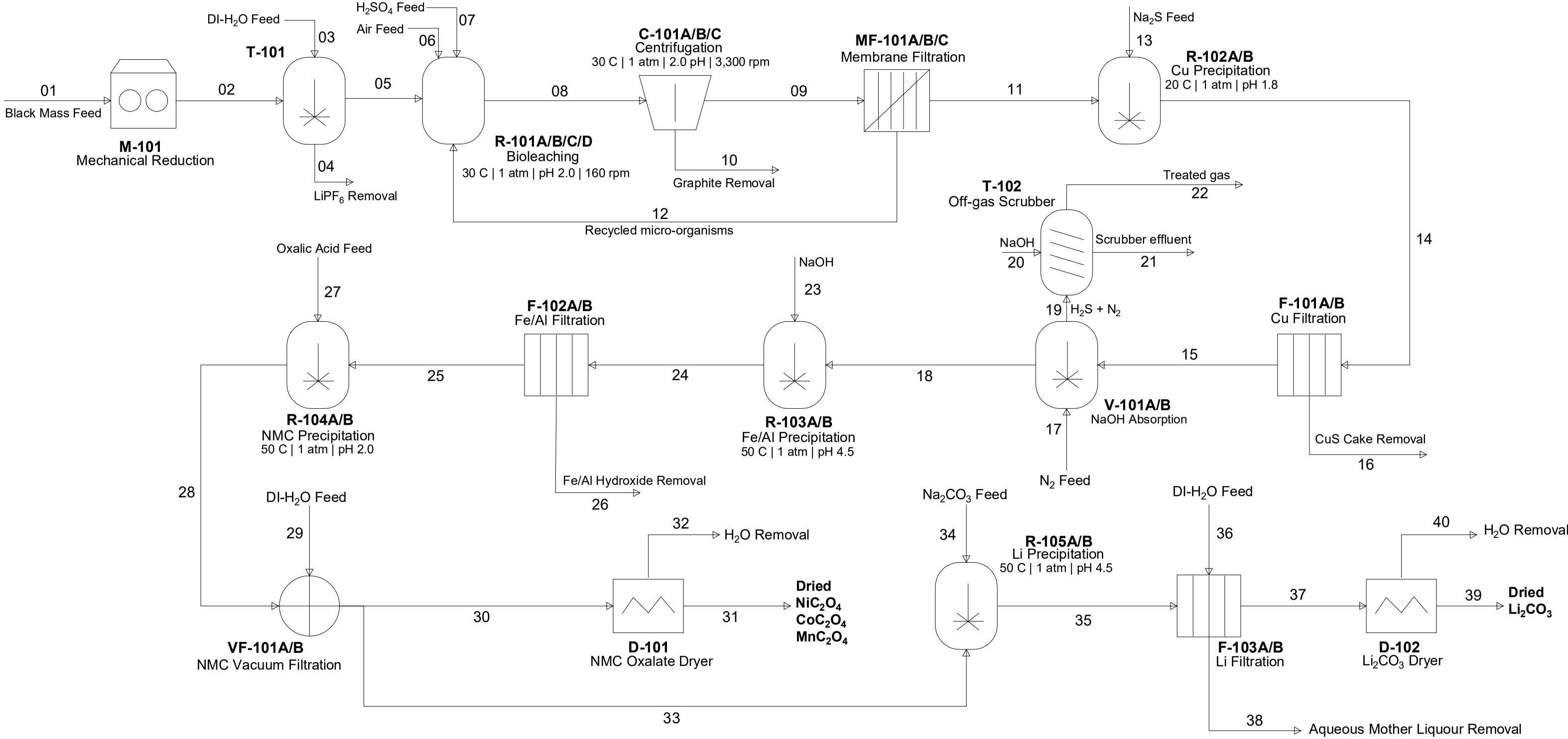

01
Black Mass Feed
M-101
Mechanical Reduction
02
T-101
DI-H2O Feed
03
04
LiPF6 Removal
05
H2SO4 Feed
07
Air Feed
06
R-101A/B/C/D
Bioleaching
30 C | 1 atm | pH 2.0 | 160 rpm
08
C-101A/B/C
Centrifugation
30 C | 1 atm | 2.0 pH | 3,300 rpm
10
Graphite Removal
09
MF-101A/B/C
Membrane Filtration
12
Recycled micro-organisms
11
Na2S Feed
13
R-102A/B
Cu Precipitation
20 C | 1 atm | pH 1.8
14
T-102
Off-gas Scrubber
Treated gas
22
NaOH
20
Scrubber effluent
21
19
H2S + N2
F-101A/B
Cu Filtration
15
CuS Cake Removal
16
V-101A/B
NaOH Absorption
17
N2 Feed
18
NaOH
23
R-103A/B
Fe/Al Precipitation
50 C | 1 atm | pH 4.5
24
F-102A/B
Fe/Al Filtration
25
Fe/Al Hydroxide Removal
26
Oxalic Acid Feed
27
R-104A/B
NMC Precipitation
50 C | 1 atm | pH 2.0
28
DI-H2O Feed
29
VF-101A/B
NMC Vacuum Filtration
30
32
H2O Removal
31
Dried
NiC2O4
CoC2O4
MnC2O4
D-101
NMC Oxalate Dryer
33
Na2CO3 Feed
34
R-105A/B
Li Precipitation
50 C | 1 atm | pH 4.5
35
DI-H2O Feed
36
F-103A/B
Li Filtration
37
38
Aqueous Mother Liquour Removal
40
H2O Removal
39
Dried
Li2CO3
D-102
Li2CO3 Dryer